

Distributed Automation Platform for Mars IoT Simulator

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Mission: Welcome to Mars. Please Don't Die.

The goal of this mission is to implement a distributed automation system capable of ingesting data from different sensors, normalize them into a Unified Event Schema and use them to enforce simple automation rules. The system should also provide a live dashboard for real-time monitoring.

The system has been implemented using the following frameworks:

- Python 3 & FastAPI, for the backend/frontend logic, asynchronous worker scripts, and REST/SSE APIs;
- RabbitMQ, as the message broker ensuring event-driven decoupling between ingestion and processing;
- MariaDB, as the relational database for the persistent storage of automation rules;
- HTML5, Vanilla JS & Bootstrap: for a lightweight and reactive frontend dashboard.
- Docker: for the containerization and orchestration of the microservices ecosystem.

User stories and requirements

Being a group of 2, we created 10 user stories:

1. As an Operator, I want to see a real-time dashboard with the latest sensor values so that I can monitor the habitat conditions.
2. As an Operator, I want to view the current status of all actuators so that I know what equipment is currently operating.
3. As an Operator, I want to manually switch actuators ON or OFF from the dashboard so that I can intervene when necessary.
4. As an Operator, I want to create an automation rule from the dashboard so that the system can react automatically to sensor changes.
5. As an Operator, I want to view the list of active automation rules so that I know what behaviors are currently configured.
6. As an Operator, I want to delete an existing automation rule so that I can remove obsolete behaviors.
7. As the System, I want to poll data from REST sensors periodically so that the latest environmental data is seamlessly fetched.
8. As the System, I want to normalize the varying payloads from different REST sensors into a unified event format so that downstream services can process them uniformly.
9. As the System, I want to evaluate incoming sensor events dynamically against persisted rules so that the system safely triggers actuators when environmental conditions change.
10. As the System, I want to persist the created automation rules in an internal database so that they survive system restarts.

LoFi Mockups

Sensors Data

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Actuators Control

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Automation Rules

Add New Rule

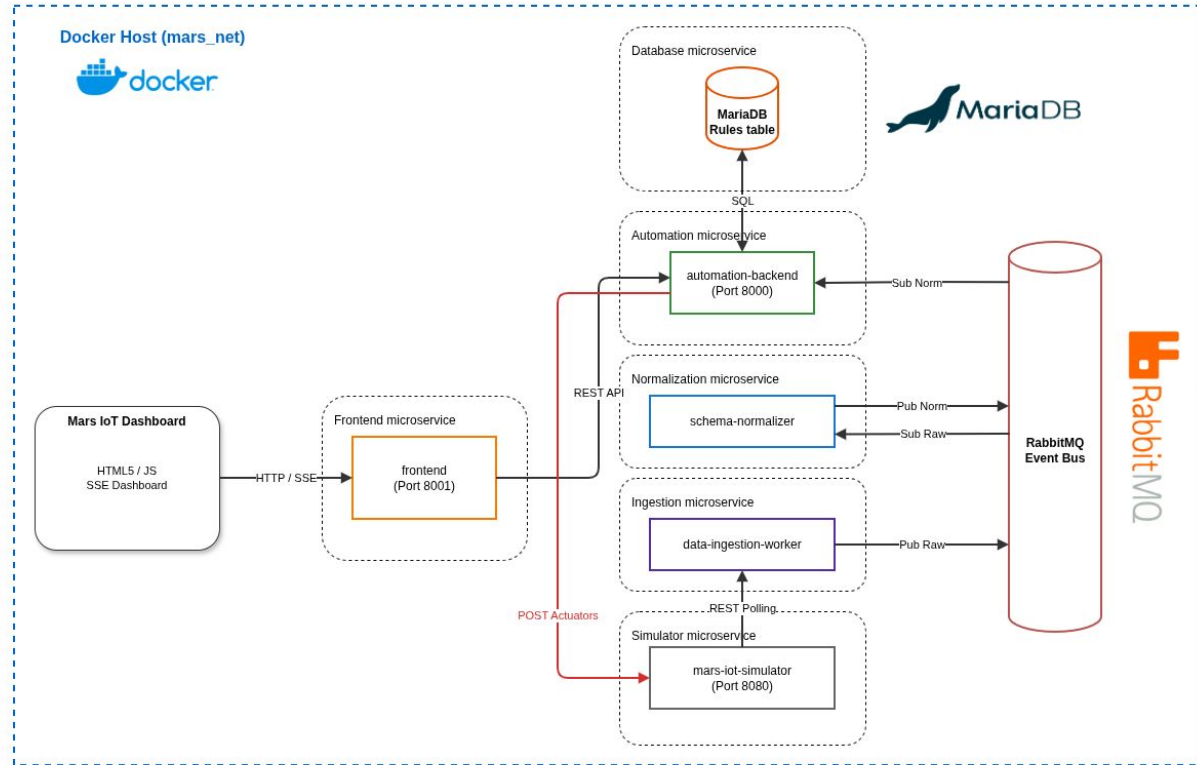
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Microservices Architecture

The entire system is containerized using Docker to ensure portability and isolation.

Our system is composed of several decoupled modules:

- Simulator: Martian IoT environment (black-box);
- Ingestion: collects raw data from the simulator;
- Normalization: responsible for data standardisation through the UES;
- Message Broker: for asynchronous communication;
- Automation Engine: enforces automation logic;
- Database: for automation rules persistence;
- Frontend: the operator dashboard.



Unified Event Schema

The secret ingredient of this system is how data normalization is performed in order to reconstruct the information brought by different sensors in a common standard:

```
{  
  "sensor_id": "string",  
  "captured_at": "string (ISO 8601 date-time)",  
  "metric": "string",  
  "value": "number",  
  "unit": "string",  
  "status": "string (ok | warning)"  
}
```

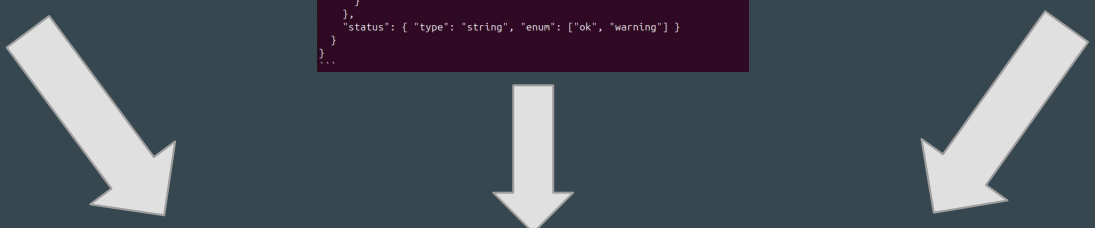
Our UES aims at processing each event as a couple "sensor-metric", as sensors with multiple metrics have to be broken down into multiple events. This allows us to generalize multi-metric sensors into scalar sensors.

Unified Event Schema (visual idea)

```
### 'rest.scalar.v1' (greenhouse_temperature, entrance_humidity, co2_hall, corridor_pressure)
'''json
{
  "type": "object",
  "required": ["sensor_id", "captured_at", "metric", "value", "unit", "status"],
  "properties": {
    "sensor_id": { "type": "string" },
    "captured_at": { "type": "string", "format": "date-time" },
    "metric": { "type": "string" },
    "value": { "type": "number" },
    "unit": { "type": "string" },
    "status": { "type": "string", "enum": ["ok", "warning"] }
  }
}
'''
```

```
### 'rest.chemistry.v1' (hydroponic_ph, air_quality_voc)
'''json
{
  "type": "object",
  "required": ["sensor_id", "captured_at", "measurements", "status"],
  "properties": {
    "sensor_id": { "type": "string" },
    "captured_at": { "type": "string", "format": "date-time" },
    "measurements": {
      "type": "array",
      "items": {
        "type": "object",
        "required": ["metric", "value", "unit"],
        "properties": {
          "metric": { "type": "string" },
          "value": { "type": "number" },
          "unit": { "type": "string" }
        }
      }
    },
    "status": { "type": "string", "enum": ["ok", "warning"] }
  }
}
'''
```

```
### 'rest.particulate.v1' (air_quality_pm25)
'''json
{
  "type": "object",
  "required": ["sensor_id", "captured_at", "pm1_ug_m3", "pm25_ug_m3", "pm10_ug_m3", "status"],
  "properties": {
    "sensor_id": { "type": "string" },
    "captured_at": { "type": "string", "format": "date-time" },
    "pm1_ug_m3": { "type": "number" },
    "pm25_ug_m3": { "type": "number" },
    "pm10_ug_m3": { "type": "number" },
    "status": { "type": "string", "enum": ["ok", "warning"] }
  }
}
'''
```



```
{
  "sensor_id": "string",
  "captured_at": "string (ISO 8601 date-time)",
  "metric": "string",
  "value": "number",
  "unit": "string",
  "status": "string (ok | warning)"
}
```

Rule Model & Logical Structure

automation_rules	
PK	id int AUTOINCREMENT
	sensor_id varchar(30) NOT NULL
	metric varchar(30) NOT NULL
	operator varchar(2) NOT NULL
	threshold_value FLOAT NOT NULL
	actuator_name varchar(30) NOT NULL
	target_state varchar(3) NOT NULL

Logical structure:

- Visual representation:
IF <Sensor> [<Metric>] <Operator> <Threshold> THEN
SET <Actuator> to <State>;
- Key blocks: Condition -> Action.

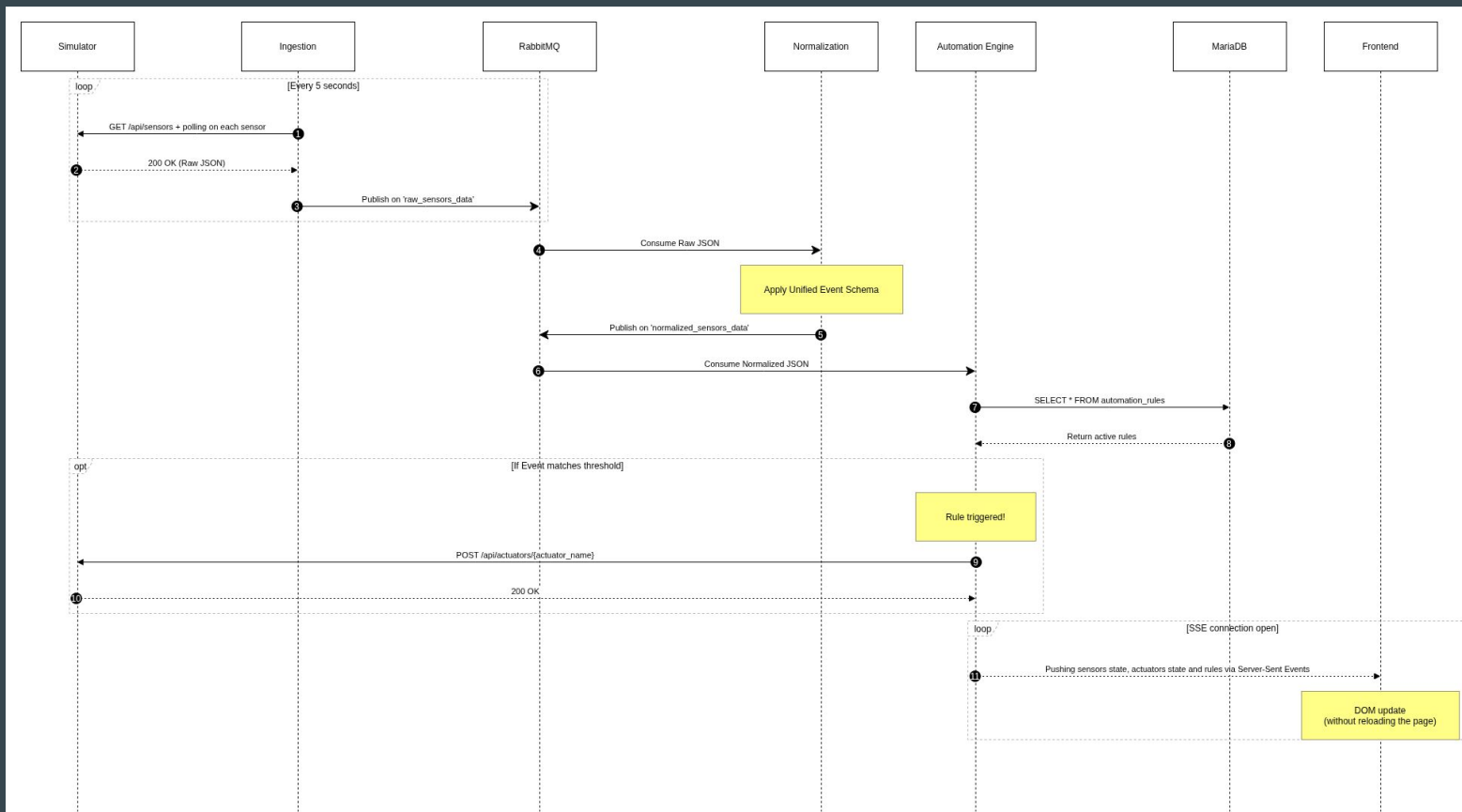
Database & persistence:

- Securely store in automation_rules table;
- Ensures automation behaviors are fully restored upon system crashes or service restarts.

System integration:

- Rules are continuously applied on the normalized data sensors;
- Active interaction with actuators APIs (e.g. turning ON the cooling_fan).

Sequence diagram



Live Demo & Q&A

The following demonstration showcases a complete docker-compose deployment, illustrating data ingestion from simulated sensors, rule evaluation in the automation engine, and real-time dashboard updates via SSE.

If you have any questions, do not hesitate to ask!

Ready to execute: docker compose up